PlatformIO Best Practices Next-generation IDE for decades, not years. Ad

Nordic nRF52840-DK

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Hardware

Platform Nordic nRF52: The nRF52 Series are built for speed to carry out increasingly complex tasks in the shortest possible time and return to sleep, conserving precious battery power. They have a Cortex-M4F processor which makes them quite capable Bluetooth Smart SoCs.

Microcontroller	NRF52840
Frequency	64MHz
Flash	1MB
RAM	256KB
Vendor	Nordic

Configuration

Please use `nrf52840_dk` ID for board option in "platformio.ini" (Project Configuration File):

```
[env:nrf52840_dk]
platform = nordicnrf52
board = nrf52840_dk
```

You can override default Nordic nRF52840-DK settings per build environment using `board_***` option, where `***` is a JSON object path from board manifest nrf52840_dk.json. For example, `board_build.mcu`, `board_build.f_cpu`, etc.

```
[env:nrf52840_dk]
platform = nordicnrf52
board = nrf52840_dk

; change microcontroller
board_build.mcu = nrf52840

; change MCU frequency
board_build.f_cpu = 64000000L
```

Uploading

Nordic nRF52840-DK supports the following uploading protocols:

- `blackmagic`
- `cmsis-dap`
- `jlink`
- `mbed`
- `nrfjprog`
- `stlink`

Default protocol is `jlink`

You can change upload protocol using upload_protocol option:

```
[env:nrf52840_dk]
platform = nordicnrf52
board = nrf52840_dk

upload_protocol = jlink
```

Debugging

Debugging - “1-click” solution for debugging with a zero configuration.

Warning

You will need to install debug tool drivers depending on your system. Please click on compatible debug tool below for the further instructions and configuration information.

You can switch between debugging Tools & Debug Probes using debug_tool option in "platformio.ini" (Project Configuration File).

Nordic nRF52840-DK has on-board debug probe and **IS READY** for debugging. You don't need to use/buy external debug probe.

Compatible Tools	On-board	Default
Black Magic Probe		
CMSIS-DAP	Yes	
J-LINK	Yes	Yes
ST-LINK		

Frameworks

Name	Description
Arduino	Arduino Wiring-based Framework allows writing cross-platform software to control devices attached to a wide range of Arduino boards to create all kinds of creative coding, interactive objects, spaces or physical experiences.
Mbed	Arm Mbed OS is a platform operating system designed for the internet of things
Zephyr	Zephyr is a new generation, scalable, optimized, secure RTOS for multiple hardware architectures